Informatica®

Informatica® Cloud Data Integration

Redis Connector

Informatica Cloud Data Integration Redis Connector
October 2025

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Preface

Use Redis Connector to learn how to read from or write to Redis by using Data Integration. Learn to create a Redis connection, run mappings in advanced mode and tasks in Data Integration.

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CHAPTER 1

Introduction to Redis Connector

You can use Redis Connector to read data from or write data to Redis.

You can use Redis Connector to read and write Redis key types HASH, LIST, and STRING. You can configure Redis Connector to read HASH and LIST keys as one row per key or as multiple rows per key to enable reading and writing Redis keys with large values.

You can use a Redis connection as a source and as a target only in mappings in advanced mode. When you run a mapping in advanced mode or a task based on a mapping in advanced mode, Data Integration analyzes the data logic in the mapping to automatically assign data logic to run on an advanced cluster.

Redis Connector assets

Create assets in Data Integration to integrate data using Redis Connector.

When you use Redis Connector, you can include the following Data Integration assets:

- Mapping
- Mapping task

Important: You can use Redis Connector only in mappings in advanced mode. You can run a task based on a mapping in advanced mode.

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

Administration of Redis Connector

Before you configure a Redis connection to connect to Redis database, you must complete the following prerequisites:

- Create a user with credentials to access the Redis server.
- Provide read and write access permissions to the user.
- If you want to use TLS for secure communication with the Redis server, enable TLS on the Redis server. Choose the TLS port and configure the TLS server certificates, key, and CA certificates.

CHAPTER 2

Connections for Redis

Use a Redis connection to access the top-level keys in Redis.

Use a Redis connection to read data from or write data to Redis database. You can create a connection on the Connections page or when you create a mapping in advanced mode or a task. Create a connection and associate it with a mapping in advanced mode. Use a Redis connection to import collections, create data objects, preview data, and run mappings in advanced mode.

When you create a Redis connection, you define the connection attributes that Data Integration uses to connect to the Redis database.

You can also parameterize a Redis connection with values specified in a parameter file when you configure a mapping in advanced mode.

Redis connection properties

When you create a Redis connection, you must configure the connection properties.

The following table describes the Redis connection properties:

Property	Description
Connection Name	The name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select which credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	Name of the runtime environment where you want to run the tasks. Select a Secure Agent or serverless runtime environment. For more information about how to configure and use the runtime environments, see *Runtime Environments* in the Administrator help.
Host	Host name or IP address of the Redis server.
Port	Redis server port number.
User	Username to access the Redis server.
Password	Password to access the Redis server.
Max Clients Per Worker	The maximum number of Redis client connections used by each worker node.
Flat Hierarchy	Enable this property to perform the following actions based on the data that you read: - Read top-level HASH keys as multiple rows with one row for each key-value pair in the hash. - Read top-level LIST keys as multiple rows with one row for each string value in the list.
Use TLS	Uses TLS to secure the communication with Redis server.
KeyStore File Path	Absolute path of the KeyStore file in the Secure Agent machine that contains private keys and certificates for the Redis server.
KeyStore Passphrase	Passphrase for the KeyStore file.

Property	Description
TrustStore File Path	Absolute path of the TrustStore file that contains certificates for the Redis server.
TrustStore Passphrase	Passphrase for the TrustStore file.

CHAPTER 3

Mappings for Redis

You can configure a mapping in advanced mode to read from and write to Redis.

Use the Data Integration Mapping Designer to define and configure a mapping in advanced mode for Redis sources and targets. In advanced mode, the Mapping Designer updates the mapping canvas to include transformations and functions that enable advanced functionality.

You can configure a Source or Target transformation to represent a Redis object. Add the Source and Target transformations in the mapping canvas and configure the Redis source and target properties. You cannot configure a filter in the Source transformation. However, you can configure a Filter transformation to filter records from the source. Configure the Filter transformation right after the Source transformation to push the filter to the Redis source.

When you run a mappings in advanced mode, Data Integration analyzes the data logic defined in the mapping to automatically assign data logic to run on an advanced cluster.

Redis sources in mappings

To read data from Redis, configure a Redis object as a Source transformation in a mapping in advanced mode.

A Redis source reads all top-level Redis keys that correspond to the Redis object name. Redis object names are converted to Redis binary-string through UTF-8 encoding. Each Redis source in a mapping is configured to read one Redis key type, that is, HASH, LIST, or STRING. Keys of other types are ignored.

You can parameterize the Redis source object with values specified in a parameter file when you configure a mapping in advanced mode.

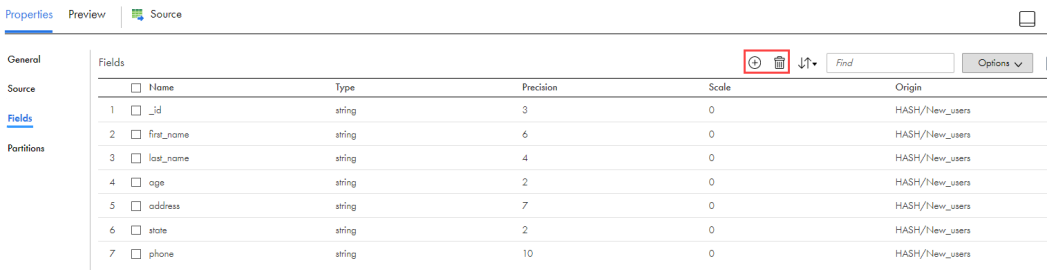
The following table describes the source properties that you can configure for a Redis source:

Property	Description
Connection	Name of the Redis source connection. You can select an existing connection or create a new connection. You can also click **New Parameter** to define a parameter for the source connection. If you want to overwrite the source connection properties at runtime, select the **Allow parameter to be overridden at run time** option. Specify the parameter file directory and name in the advanced session properties.
Source Type	Type of the source object. Select Single Object.
Object	Name of the Redis source object based on the selected source type. Select the source object for a single source.

Adding and removing source fields

You can configure the source fields that you want to use in the data flow.

You can add or remove source fields of primitive data types. You can add fields to a mapping source and retrieve a field from the source object that is not displayed in the list.



For more information on how to add or remove source fields, see the topic "Add and remove fields" in Source transformation in the Data Integration documentation.

Note: You cannot add or remove source fields for Redis STRING data type. Otherwise, the mapping throws a runtime exception.

Redis targets in mappings

To write data to a Redis target, configure a Redis object as a Target transformation in a mapping in advanced mode.

A Redis target writes incoming rows as top-level Redis keys. Each Redis target in a mapping is configured to write all rows as one Redis key type HASH, LIST, or STRING. Redis Connector creates keys with names in ${OBJECT_NAME}:${ID_COLUMN_VALUE} format. Redis object names are converted to Redis binary-string through UTF-8 encoding.

You can parameterize the Redis target object with values specified in a parameter file when you configure a mapping in advanced mode.

The following table describes the target properties that you can configure for a Redis target:

Property	Description
Connection	Name of the Redis connection that is associated with a dataset. You can select an existing connection or create a new connection. You can also click **New Parameter** to define a parameter for the target connection. To overwrite the target connection properties at runtime, select the **Allow parameter to be overridden at run time** option.
Target Type	Type of target object. Select Single Object or Parameter. You can add multiple Redis targets in a mapping in advanced mode.
Object	Name of the Redis target object based on the target type selected. You can select an existing target object or create a new target object at runtime.
Create New at Runtime	Creates a Redis target table at runtime based on the target table you specify. To create a target table at runtime, specify the following details: - Enter the Redis object name to which you want to write in the **Object Name** field. When you write a row to a Redis object, the name of the top-level Redis key is in ${OBJECT_NAME}:${ID_COLUMN_VALUE} format. - Enter the row type in the **Path** field. You can specify row types, HASH, LIST, or STRING based on the row type in the target object. Row types must be in uppercase. The advanced cluster inserts Redis keys into a Redis database that does not already have Redis keys that match the target key pattern. You can edit the metadata of the source fields before creating the target.
Operation	The target operation. You can select the **Insert** operation. Other operations are not applicable.

Redis key types

Redis is a key-value store that contains a list of keys and values. The value of each key is a particular type, including HASH, LIST, and STRING.

Redis Connector supports HASH, LIST, and STRING key types. The supported key types can have the following values:

- HASH value type is a list of binary-string-key and binary-string-value pairs.
- LIST value type is a list of binary-string values.
- STRING value type is a single binary-string value.

The fields in Redis sources match the corresponding schema as shown in the following table. For Redis targets, the mapped incoming fields must exactly match the corresponding schema as shown in the following table:

Key types	If flat hierarchy is not enabled	If flat hierarchy is enabled
HASH	Schema at source is inferred from sampling the data object. Schema at target is inferred from mapped incoming fields. Targets must have an "_id" field.	_id string hkey string value string
LIST	_id string value array of string	_id string value string
STRING	_id string value string	_id string value string

You can use an Expression or Hierarchy Processor transformation to obtain the required schema for the target.

Edit metadata in a Redis source and target

You can edit data types to customize the precision, scale, and data types of fields. You can edit the field metadata in a Source or Target transformation to change data types, precision, and scale. To edit metadata, use the **Edit Metadata** option for the source and target fields on the **Fields** tab of the Properties panel.

Reading and writing hierarchical data in mappings

You can use the Hierarchy Processor transformation in a mapping in advanced mode to process data from complex data sources. You can read complex data types with hierarchical input and write as relational output. You can also read complex data types with relational input and write as hierarchical output.

When you configure a mapping in advanced mode to convert relational to hierarchical output, you can join data sources, group by and order by data fields, filter for specific information, and aggregate both the input and output data. When you configure a mapping in advanced mode to convert hierarchical to relational output, you can read from a single hierarchical data source and write to multiple relational data targets.

For more information about configuring a Hierarchy Processor transformation, see the Hierarchy Processor transformation section in *Transformations* in Data Integration help.

Partitioning

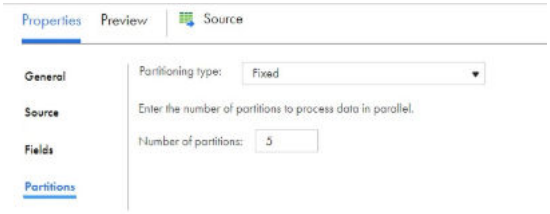
When you read data from Redis, you can configure partitioning to optimize the mapping performance at run time. The partition type controls how the agent distributes data among partitions at partition points.

Fixed partitioning for Redis sources

You can configure fixed partitioning in a mapping that reads data from Redis sources.

You can enable partitioning when you configure the Source transformation in the Mapping Designer.

On the **Partitions** tab for the Source transformation, you select fixed partitioning and enter the number of partitions, as shown in the following image:



The mapping task distributes rows of data based on the number of partitions that you specify.

Rules and guidelines for Redis mappings

Consider the following guidelines before you run a mapping in advanced mode:

- Redis Connector samples rows to populate the object catalog. Use the search field to search for a specific object name if the source object is not displayed in the list.
- The source tables must not be empty. Otherwise, the mapping fails.
- All rows belonging to a HASH-type Redis source must have the same schema. Otherwise, the schema inference might result in incorrect results.
- If the source contains binary fields, the Data Preview section shows jumbled text.

CHAPTER 4

Data type reference

Data Integration uses the following data types in mappings with Redis:

Redis native data types

Binary-string is the only native field data type of Redis. The Redis binary-string data type appears in the **Fields** tab for Source and Target transformations when you choose to edit metadata for the fields.

Transformation data types

Set of data types that appear in the transformations. They are internal data types based on ANSI SQL-92 generic data types, which the Secure Agent uses to move data across platforms. Transformation data types appear in all transformations in a mapping.

When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

Redis and transformation data types

The Redis native type is binary-string for all supported Redis simple values. A Redis binary-string is a variable length array of bytes. You can map Redis binary-string type to the transformation data types, binary and string. Redis binary-string values are converted to and from transformation string data type values through UTF-8 encoding.

The following table describes the Redis data types that Data Integration supports and the corresponding transformation data types:

Redis Data Types	Transformation Data Types	Range and Description
String	Binary	Null values cause row errors in the target.
String	String	Null values cause row errors in the target.

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